

Phosphate plant -- Acid

Section report: Acid

Site	Atlas Phosphate Complex (:atlas_phosphate)
Reporting period	2026-01-01T00:00:00 to 2026-12-31T23:00:00 (365 days, 8568 operating hours)
Product	Dap from 29.0 % P2O5 ore, 1713.0 kt of product
Framework scope	ISO 50001:2018 energy management; ISO 14001:2015 environmental management; ISO 46001:2019 water efficiency; ISO 9001:2015 quality management; GHG Protocol corporate standard; EU Industrial Emissions Directive 2010/75; US EPA 40 CFR 63 subpart AA; IFA/EFMA BAT for phosphate fertilisers; Phosphogypsum stack management code
Provenance	Design revision A seed 20260101 8760 hourly intervals of 69 signals generated 2026-09-23T15:43:11.110 basis :screening_plant_model

Phosphoric acid: balance, specification and cost

The acid plant produced 446 kt of merchant acid -- 249.1 kt P2O5, 343.9 kt H3PO4 on the 100 % basis -- at 55.85 % P2O5, which is the :technical grade. The evaporators removed 1,574.8 thousand t of water with 0.41149 t of steam per t of P2O5 concentrated, an economy of 3.76 t of water per t of steam. The acid leaves the plant Compliant against the specification of the traded grade.

THE MERCHANT ACID OF THE CAMPAIGN

Quantity	Value
Merchant acid	446 kt (249.1 kt P2O5, 343.9 kt as H3PO4)
Grade	55.85 % P2O5 (77.12 % H3PO4) -- :technical
P2O5 of the attack	the filters receive 24.5 % of it as merchant acid, the rest goes to the granulation line
Yield	the filtration and the attack keep 97.68 % of the P2O5 fed; the evaporation closes at 101.96 %
Concentration	2.82 m3 of feed and 1.14 GJ of steam per t P2O5, 61 kWh of blower and pumps
Cost	503 :USD per t P2O5 as acid (281.1 per t of acid) against a price of 905 and a margin of 402 per t P2O5

THE MERCHANT ACID AGAINST THE SPECIFICATION OF THE TRADED GRADE

Specification	Instrument	Campaign	Limit	Unit	Direction	Margin %	Status
Colour	:ACID_COLOUR		2.86	4 :count	:le	28.5	Compliant
F	:ACID_F		0.3533	0.5 :wt_pct	:le	29.35	Compliant
Fe al	:ACID_FE_AL		0.1918	1 :wt_pct	:le	80.82	Compliant
P2o5	:STRONG_ACID_P2O5		55.85	52 :wt_pct	:ge	7.41	Compliant
So4	:ACID_SO4		1.16	1.5 :wt_pct	:le	22.39	Compliant
Solids	:ACID_SOLIDS		0.2943	0.5 :wt_pct	:le	41.13	Compliant

WHERE THE P2O5 OF THE ATTACK GOES

Destination	t P2O5	Share %	Note
Merchant acid	249,077	23.9	to the acid storage, the grade the customer buys
Fertiliser route	768,813	73.78	to the granulation line as concentrated acid
Gypsum cake	24,161	2.32	water-soluble P2O5 that left with the phosphogypsum
Unaccounted	0	0	sampling and rounding of the balance, targets zero

THE STEPS OF THE ACID ROUTE

Step	In (t P2O5)	Out (t P2O5)	Loss (t P2O5)	Yield %	Note
Attack and filtration	1.042052e6	1.017891e6	24,161	97.68	the cake carries the free P2O5 out of the circuit
Concentration	1.017891e6	244,294	0	101.96	the closure of the evaporation against the registered merchant split
Merchant acid	244,294	249,077	0	101.96	the 24.5 % of the filtered P2O5 the storage tank receives

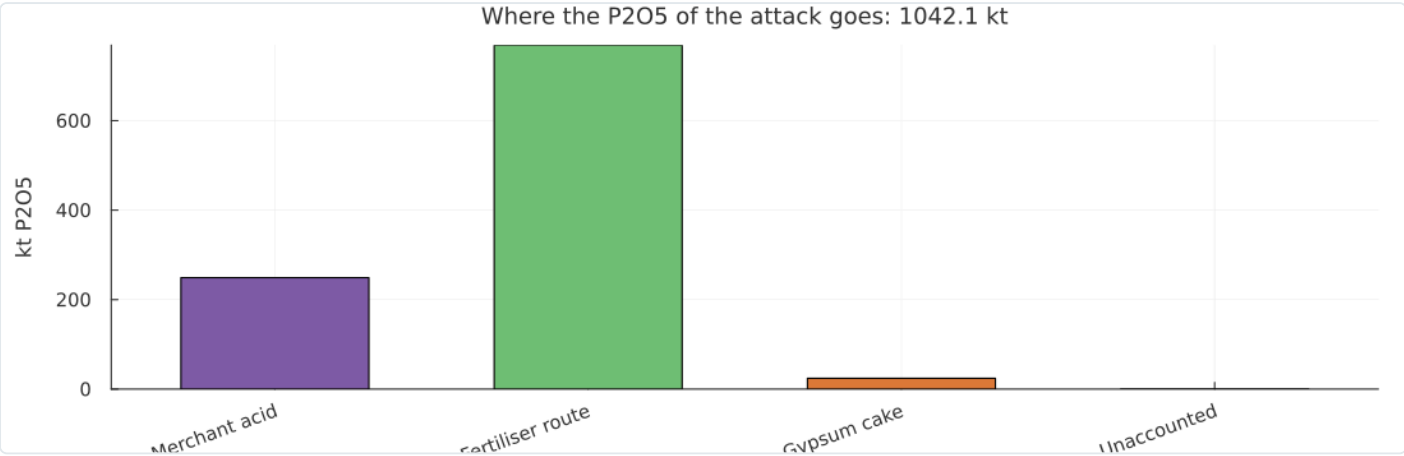
WHAT A TONNE OF P2O5 COSTS IN ROCK, ACID, STEAM AND WATER

Consumption	Value	Unit	Basis	Note
Rock	3.22	:t_ph	:per_p2o5_sold	t of rock per t P2O5 sold as acid, allocated by the acid share
Sulphuric acid	2.89	:t_ph	:per_p2o5_sold	t of H2SO4 per t P2O5 sold as acid
Sulphur	0.9446	:t_ph	:per_p2o5_sold	t of sulphur burnt per t P2O5 sold as acid
Steam	0.4115	:t_ph	:per_p2o5_concentrated	t of steam per t P2O5 the evaporators concentrate, both products included
Electricity	60.82	:kwh_per_t	:per_p2o5_concentrated	kWh per t P2O5 concentrated: blower of the acid plant and evaporator set
Water	2.82	:m3_ph	:per_p2o5_concentrated	m3 of evaporator feed per t P2O5 concentrated
Thermal energy	1.14	:gj_per_t	:per_p2o5_concentrated	GJ of steam per t P2O5 concentrated
Gypsum	4.76	:t_ph	:per_p2o5_sold	t of phosphogypsum per t P2O5

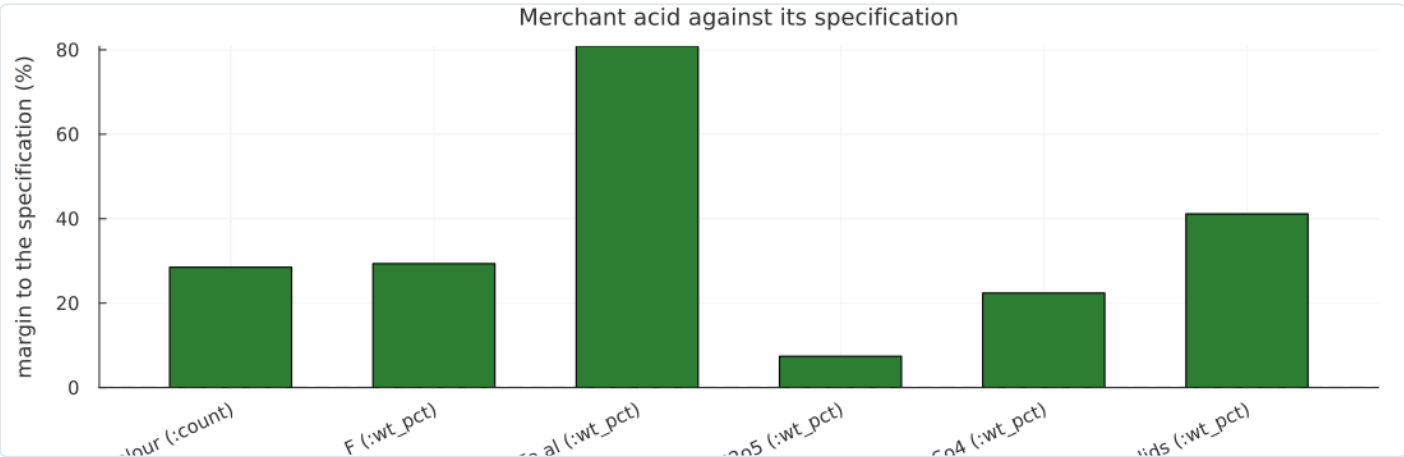
THE TARGETS OF THE ACID ROUTE

Metric	Campaign	Target	Unit	Direction	Margin %	Status
:acid_concentration_yield		101.96	97 :pct	:ge	5.11	Compliant
:acid_solids		0.2943	0.5 :wt_pct	:le	41.13	Compliant
:acid_electricity		60.82	75 :kwh_per_t	:le	18.9	Compliant
:acid_so4		1.16	1.5 :wt_pct	:le	22.39	Compliant
:acid_steam		0.4115	0.62 :t_ph	:le	33.63	Compliant

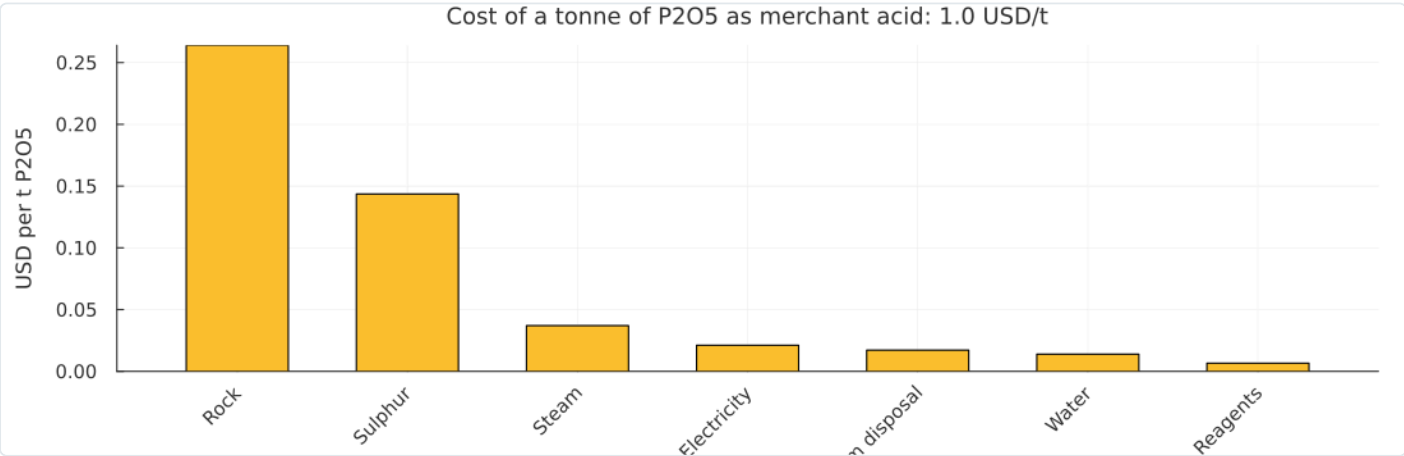
Metric	Campaign	Target	Unit	Direction	Margin %	Status
acid_cost		503.33	600 :usd_per_t	:le	16.11	Compliant
acid_f		0.3533	0.5 :wt_pct	:le	29.35	Compliant
acid_p2o5_yield		97.68	96.5 :pct	:ge	1.22	Compliant
acid_strength		55.85	52 :wt_pct	:ge	7.41	Compliant



Where the P2O5 of the attack goes: the acid route and the fertiliser route



The merchant acid against the specification of the traded grade



Cost of a tonne of P2O5 as merchant acid, item by item

Acid route: the plant kept 97.68 % of the P2O5 at the filters, sold 446 kt of acid at 55.85 % P2O5 for 905 per t P2O5, and spent 503.